FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

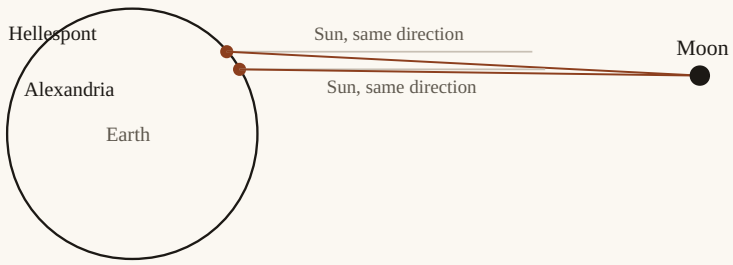
# Hipparchus: the Moon changes place when we move

Hold a finger at arm's length and look with one eye, then the other. The finger jumps against the far wall. The gap between your eyes is the baseline. The angle of the jump, with that baseline, is a distance. That is parallax. Hipparchus, in the second century BCE, used it on the Moon. Most of his books are lost. Later astronomers rebuilt the argument from what they still had.

One famous piece uses a solar eclipse. Near the Hellespont the Sun's disk was totally obscured. At Alexandria four-fifths of that disk was obscured, so only about one-fifth of the Sun's diameter stayed uncovered. The whole shift between the two places was that fifth of the diameter. It is a fraction of the diameter, not of the area: four-fifths of the area would be a different, larger bite. A date often proposed is 14 March 190 BCE. The identification is not secure, and this lesson will not pretend it is. Hipparchus used reports. He did not stand in both cities.

Two places on the Earth, one Moon, parallel sunlight

Stations are on the circle. Grey lines are parallel. The Moon is drawn much too close.




Both dots are on the circle. Hellespont is north of Alexandria. The grey lines are parallel. The Moon is drawn much too close, so the angle is enlarged. Total at the Hellespont. At Alexandria four-fifths of the diameter was covered, so about one-fifth stayed bright.

Two stations, one Moon, a shared solar direction

The angle is drawn far too large. The Sun's own parallax is treated as negligible beside the Moon's. Drawn for this lesson.

The baseline is the north-south distance on the Earth between those latitudes. The sightlines to the Moon differ. The Sun is so far that its direction can be treated as the shared wall. G. J. Toomer's reading of the reconstruction gives one lower bound near 71 Earth radii. Other ways of treating the shadow and the Sun's distance give a band roughly 59 to 67 Earth radii. That band is not a modern confidence interval. The Moon really averages near 60 Earth radii, and the distance changes a little through the month. An ancient model can land close to the truth and still be a model with wide edges. Nearness is not the same thing as a tight measurement.

Compare the sticks. The Venus stations of 1761 and 1769 stand on the Earth's surface, a baseline of hundreds or thousands of miles. Kepler's Mars triangles, later in this day's lessons, use a baseline the size of Earth's orbit. Hipparchus is the same idea with the cities of two reports. Aristarchus, just before this page, was comparing sizes inside one triangle. Here the observer moves, and the Moon moves on the sky.

The illustrated lesson is download the PDF.

## Sources

- MacTutor, biography of Hipparchus: https://mathshistory.st-andrews.ac.uk/Biographies/Hipparchus/
- MacTutor, Dictionary of Scientific Biography excerpt: https://mathshistory.st-andrews.ac.uk/DSB/Hipparchus.pdf

## Key facts

- Parallax is a known baseline and an angle. Hipparchus's own books are mostly lost and were rebuilt by later writers.
- The eclipse was total at the Hellespont. At Alexandria four-fifths of the Sun's diameter was covered, so about one-fifth of the diameter stayed bright. That fifth is the parallax shift. It is not four-fifths of the area. He used reports. The 190 BCE date is uncertain.
- Reconstructions give a lower bound near 71 Earth radii in one reading, and roughly 59 to 67 in others. That is not one confidence interval.
- The modern mean is near 60 Earth radii and varies. Venus transits use a surface baseline. Kepler's Mars work uses the orbit.